Year 7 Mathematics

Unit 4: Fractions

4.4 + 4.5 Multiplying by a Fraction

Multiplying by a Fraction

STARTER · 5 MIN

Date: _____ **Title: Multiplying by a Fraction**

LEARNING OBJECTIVES

- To be able to multiply a fraction by a whole number
- To be able to multiply a fraction by a fraction

STARTER *Answer in your book. No calculators.*

1. $4 \times 5 = 5 \times \square$
2. $3 \times 8 = \square \times 3$
3. $\frac{1}{4}$ of £120
4. $\frac{2}{3}$ of 18 cm

Starter — Answers

STARTER

Tick or correct in green pen

1. **4**

2. **8**

3. **£30**

4. **12 cm**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find a fraction of an amount and multiply a whole number by a fraction	☐
SILVER	I can multiply a fraction by a fraction	☐
GOLD	I can multiply mixed numbers and solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Multiplying by a Fraction

INTRODUCTION · 3 MIN

'Of' means multiply. To multiply fractions, multiply the tops and multiply the bottoms.

Fraction of

÷ by the bottom, × by the top.

$\frac{2}{3}$ of 12 m = 12 ÷ 3 × 2 = 8 m

Whole × fraction

Same as the fraction of it.

$10 \times \frac{2}{5} = \frac{2}{5}$ of 10 = 4

Fraction × fraction

Tops × tops, bottoms × bottoms.

$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$

Mixed number

Make it improper first.

$1\ \frac{1}{2} = \frac{3}{2}$

Examples 1 and 2 — A Fraction of an Amount

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint ÷ bottom, then × top Whole × fraction = fraction of

Example 1

Work out

$\frac{2}{3}$ of 12 m

Example 2

Work out

$10 \times \frac{2}{5}$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

÷ by the bottom, × by the top

12 ÷ 3 = 4 m

4 × 2 = 8 m

Tip: one third first, then two thirds.

Example 2

10 × two fifths = two fifths of 10

10 ÷ 5 = 2

2 × 2 = 4

Tip: × and 'of' give the same answer.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out*

Hint ÷ bottom, then × top Whole × fraction = fraction of

1 $\frac{4}{5}$ of 30

2 $\frac{2}{3}$ of 90

3 $\frac{2}{7}$ of 42

4 $\frac{3}{10}$ of 10

5 $45 \times \frac{1}{9}$

★ $16 \times \frac{7}{8}$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **24**

2 **60**

3 **12**

4 **3**

5 **5**

★ **14**

Examples 3 and 4 — Multiplying Two Fractions

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Tops × tops Bottoms × bottoms

Example 3

Work out

$$\frac{3}{4} \times \frac{1}{2}$$

Example 4

Work out

$$\frac{2}{3} \times \frac{2}{3}$$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Tops × tops, bottoms × bottoms

$$\frac{3}{4} \times \frac{1}{2} = \frac{3 \times 1}{4 \times 2}$$

$$= \mathbf{\frac{3}{8}}$$

Tip: three quarters of a half.

Example 4

Tops × tops, bottoms × bottoms

$$\frac{2}{3} \times \frac{2}{3} = \frac{2 \times 2}{3 \times 3}$$

$$= \mathbf{\frac{4}{9}}$$

Tip: the answer is smaller than both.

Task 2

PRACTICE · 9 MIN

SILVER *Work out. Simplify where you can*

Hint Tops × tops Bottoms × bottoms

1 $\frac{1}{2} \times \frac{1}{3}$

2 $\frac{1}{4} \times \frac{2}{3}$

3 $\frac{2}{5} \times \frac{1}{2}$

4 $\frac{3}{5} \times \frac{2}{7}$

5 $\frac{3}{4} \times \frac{3}{4}$

★ Area of a square of side $\frac{3}{4}$ cm

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $\frac{1}{6}$

2 $\frac{2}{12} = \frac{1}{6}$

3 $\frac{2}{10} = \frac{1}{5}$

4 $\frac{6}{35}$

5 $\frac{9}{16}$

★ $\frac{9}{16}$ cm²

Examples 5 and 6 — Mixed Numbers and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Mixed: make it improper 'Of' means ×

Example 5

(2 marks)

Work out

$3\frac{1}{2} \times 5$

Example 6

(2 marks)

$\frac{1}{20}$ of a council are from minority ethnic groups.

$\frac{1}{3}$ of these are women.

What fraction of the council are minority ethnic women?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Make it an improper fraction

$$3\frac{1}{2} = \frac{7}{2}$$

$$\frac{7}{2} \times 5 = \frac{35}{2} = 17\frac{1}{2}$$

Tip: or $3 \times 5 + \frac{1}{2} \times 5 = 15 + 2\frac{1}{2}$.

Example 6

'Of' means multiply

$$\frac{1}{3} \times \frac{1}{20} = \frac{1}{60}$$

Tip: $3 \times 20 = 60$.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: give answers in simplest form

Hint Mixed: make it improper 'Of' means ×

1 $2\frac{1}{3} \times 6$

2 $1\frac{3}{4} \times 8$

3 $1\frac{1}{2} \times \frac{2}{3}$

4 Rectangle: $\frac{5}{6}$ m by 4 m. Area?

5 $\frac{1}{2}$ off, then $\frac{1}{3}$ off the sale price. Final fraction?

★ Square of side $1\frac{1}{2}$ m. Area?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1 14

2 14

3 1

4 $\frac{20}{6} = 3\ \frac{1}{3}\ m^2$

5 $\frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$

★ $\frac{9}{4} = 2\ \frac{1}{4}\ m^2$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out $\frac{3}{5}$ of 40.

SILVER Work out $\frac{2}{3} \times \frac{4}{5}$.

GOLD Work out $2\frac{1}{2} \times 3$.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE 24

SILVER $\frac{8}{15}$

GOLD $\frac{5}{2} \times 3 = \frac{15}{2} = 7\frac{1}{2}$

Exam Practice — Fractions of Amounts

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
÷ bottom, then × top
Whole × fraction = fraction of

Question 1

(2 marks)

Work out

(a) $\frac{1}{4}$ of £120 (1)

..... £

(b) $\frac{5}{9}$ of 36 kg (1)

..... kg

Question 2

(2 marks)

Work out

(a) $22 \times \frac{4}{11}$ (1)

.....

(b) $10 \times \frac{3}{5}$ (1)

.....

Exam Practice — Multiplying Fractions

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Tops × tops Bottoms × bottoms

Question 3

(2 marks)

Work out

(a) $\frac{1}{4} \times \frac{2}{3}$ (1)

.....

(b) $\frac{2}{5} \times \frac{1}{2}$ (1)

.....

Question 4

(2 marks)

A square has sides of $\frac{3}{4}$ cm.

Work out its area.

..... cm^2

Exam Practice — Mixed Numbers

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Mixed: make it improper 'Of' means ×

Question 5

(2 marks)

Work out

$4\frac{2}{3} \times 6$

.....

Question 6

(3 marks)

A rectangle is $2\frac{1}{2}$ m by $\frac{2}{3}$ m.

Work out its area.

Give your answer as a mixed number.

..... m^2

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) £30 — B1

(b) 20 kg — B1

Tip: 36 ÷ 9 × 5.

Question 2

(a) 8 — B1

(b) 6 — B1

Tip: 22 ÷ 11 × 4.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $\frac{2}{12} = \frac{1}{6}$ — B1

(b) $\frac{2}{10} = \frac{1}{5}$ — B1

Tip: simplify your answer.

Question 4

$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$ cm² — M1 A1

Tip: area of a square = side × side.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$$\frac{14}{3} \times 6 = \frac{84}{3}$$ M1

$$= 28$$ A1

Tip: or $4 \times 6 + \frac{2}{3} \times 6 = 24 + 4$.

Question 6

$$\frac{5}{2} \times \frac{2}{3} = \frac{10}{6}$$ M1 M1

$$= 1\ \frac{2}{3}\ \text{m}^2$$ A1

Tip: make the mixed number improper first.